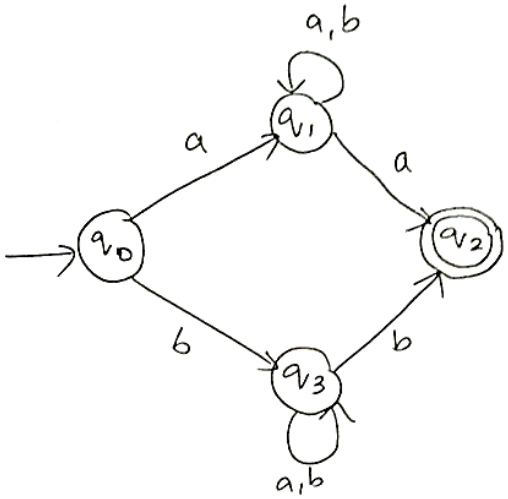
SAVEETHA SCHOOL OF ENGINEERING

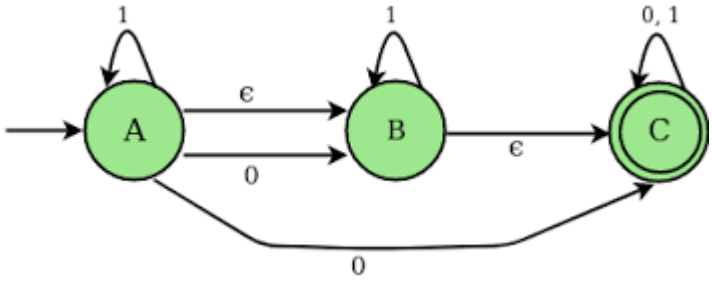
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING

CSA13 – THEORY OF COMPUTATION

ASSIGNMENT – 1

		CO	BL
1	Design a Deterministic Finite Automata (DFA) to accept strings that start and end with same symbols over **Σ**={0,1}. Write the formal definition of the DFA. Also show that the string **010100** is accepted by the DFA	1	3
2	Design NFA to accept string having 101 as substring	1	3
3	Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true. $$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$	1	1
4	Consider the NFA given below which accepts the language representing the set of all strings that start and end with the same symbol over the alphabet **Σ**={a,b}. Construct a DFA that represents the same language 	1	3
5	A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states	1	3



ANSWERS:

1. Language Description:

The language consists of strings over $\Sigma = \{0,1\}$ that start and end with the same symbol (e.g., strings of length ≥ 1 like $0,1,00,11,010,101,010100$ - wait, let's check 010100: starts with 0, ends with 0, so it should be accepted).

State Description:

- q_0: Initial state (before reading any input symbol).
- q_1: State after reading starting symbol 0.
- q_2: State after reading starting symbol 1.
- q_3: State after reading a string starting with 0 and ending with 1.
- q_4: State after reading a string starting with 1 and ending with 0.

Transition Table:

Present State	Next State for Input 0	Next State for Input 1
$\rightarrow q_0$ (Start)	q_1	q_2
$* q_1$ (Final)	q_1	q_3
$* q_2$ (Final)	q_4	q_2
q_3	q_1	q_3
q_4	q_4	q_2

(Note: $ q_1$ and $* q_2$ are final/accepting states).*

Formal Definition of DFA:

$M = (Q, \Sigma, \delta, q_0, F)$ where:

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- q_0 is the initial state
- $F = \{q_1, q_2\}$ (Accepting states)
- δ is the transition function defined as:
    - $\delta(q_0, 0) = q_1, \delta(q_0, 1) = q_2$
    - $\delta(q_1, 0) = q_1, \delta(q_1, 1) = q_3$
    - $\delta(q_2, 0) = q_4, \delta(q_2, 1) = q_2$
    - $\delta(q_3, 0) = q_1, \delta(q_3, 1) = q_3$
    - $\delta(q_4, 0) = q_4, \delta(q_4, 1) = q_2$

Simulation for string 010100:

- $\hat{\delta}(q_0, 010100) = \hat{\delta}(\delta(q_0, 0), 10100) = \hat{\delta}(q_1, 10100)$
- $= \hat{\delta}(\delta(q_1, 1), 0100) = \hat{\delta}(q_3, 0100)$
- $= \hat{\delta}(\delta(q_3, 0), 100) = \hat{\delta}(q_1, 100)$
- $= \hat{\delta}(\delta(q_1, 1), 00) = \hat{\delta}(q_3, 00)$
- $= \hat{\delta}(\delta(q_3, 0), 0) = \hat{\delta}(q_1, 0)$
- $= \delta(q_1, 0) = q_1$

Since $q_1 \in F$, the string 010100 **is accepted** by the DFA.

2.

Language Description:

The language accepts any string over $\Sigma = \{0,1\}$ that contains 101 anywhere within it (i.e., of the form $(0 + 1)^* 101 (0 + 1)^*$).

Transition States:

- Let q_0 be the initial state. The machine stays in q_0 on inputs 0 or 1 until it encounters the first 1.
- Upon reading 1, it moves to state q_1.
- From q_1, on reading 0, it moves to state q_2.
- From q_2, on reading 1, it moves to state q_3 (which is the accepting state).
- Once in the final state q_3, it can accept any subsequent characters 0 or 1 (self-loops on q_3).

Transition Table:

Present State	Input 0	Input 1
$\rightarrow q_0$	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	ϕ
$* q_3$	$\{q_3\}$	$\{q_3\}$

3. Step 1: Base Case (For $n = 1$)

- **LHS:** $\sum_{i=1}^{1} i\,(i+1) = 1(1+1) = 1(2) = 2$
- **RHS:** $\frac{1(1+1)(1+2)}{3} = \frac{1(2)(3)}{3} = \frac{6}{3} = 2$

Since LHS $=$ RHS, the statement is true for $n = 1$.

Step 2: Induction Hypothesis

Assume that the statement is true for some positive integer $n = k$. That is:

$$\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$$

Step 3: Inductive Step (For $n = k + 1$)

We need to prove that the statement holds for $n = k + 1$, i.e.:

$$\sum_{i=1}^{k+1} i\,(i+1) = \frac{(k+1)(k+2)(k+3)}{3}$$

Consider the LHS for $n = k + 1$:

$$\sum_{i=1}^{k+1} i\,(i+1) = \left(\sum_{i=1}^{k} i\,(i+1)\right) + (k+1)(k+2)$$

Using the induction hypothesis:

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

Take $(k+1)(k+2)$ common out:

$$= (k+1)(k+2)\left[\frac{k}{3}+1\right]$$

$$= (k+1)(k+2)\left[\frac{k+3}{3}\right]$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

This matches the RHS for $n = k+1$.

Hence, by the principle of mathematical induction, the formula is true for all positive integers n. **Q.E.D.**

4. Given NFA Analysis:

- States: $Q = \{q_0, q_1, q_2, q_3\}$
- Initial State: q_0
- Final State: q_2 (assuming from the double circle in the diagram)
- Transitions:
  - $\delta(q_0, a) = \{q_1\}$, $\delta(q_0, b) = \{q_3\}$
  - $\delta(q_1, a) = \{q_1, q_2\}$, $\delta(q_1, b) = \{q_1\}$ — *Wait, looking closely at the diagram:*
    - From q_0 on $a \to q_1$, on $b \to q_3$
    - From q_1 on $a, b \to q_1$ (loop), and on $a \to q_2$
    - From q_3 on $a, b \to q_3$ (loop), and on $b \to q_2$
    - q_2 is final.

Subset Construction Method:

1. **Initial State of DFA:** $[q_0]$
2. **Transitions for** $[q_0]$**:**
   - $\delta'([q_0], a) = \delta(q_0, a) = \{q_1\} \to$ Let's call this state A
   - $\delta'([q_0], b) = \delta(q_0, b) = \{q_3\} \to$ Let's call this state B
3. **Transitions for State** $\{q_1\}$ **(starts with** a**):**
   - On a: $\delta(q_1, a) = \{q_1, q_2\} \to$ Let's call this state C (Accepting since it contains q_2)
   - On b: $\delta(q_1, b) = \{q_1\} \to$ State A
4. **Transitions for State** $\{q_3\}$ **(starts with** b**):**
   - On a: $\delta(q_3, a) = \{q_3\} \to$ State B

- On b: $\delta(q_3, b) = \{q_2, q_3\}$ → Let's call this state D (Accepting since it contains q_2)

5. **Transitions for State $\{q_1, q_2\}$ (C):**
   - On a: $\delta(\{q_1, q_2\}, a) = \delta(q_1, a) \cup \delta(q_2, a) = \{q_1, q_2\} = C$
   - On b: $\delta(\{q_1, q_2\}, b) = \delta(q_1, b) \cup \delta(q_2, b) = \{q_1\} = A$
6. **Transitions for State $\{q_2, q_3\}$ (D):**
   - On a: $\delta(\{q_2, q_3\}, a) = \delta(q_2, a) \cup \delta(q_3, a) = \{q_3\} = B$
   - On b: $\delta(\{q_2, q_3\}, b) = \delta(q_2, b) \cup \delta(q_3, b) = \{q_2, q_3\} = D$

DFA Transition Table:

DFA States	Input a	Input b
$\rightarrow q_0$	$\{q_1\}$	$\{q_3\}$
$\{q_1\}$	$\{q_1, q_2\}$	$\{q_1\}$
$\{q_3\}$	$\{q_3\}$	$\{q_2, q_3\}$
$*\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_1\}$
$*\{q_2, q_3\}$	$\{q_3\}$	$\{q_2, q_3\}$

(Where $*$\{q_1, q_2\}$ and $*\{q_2, q_3\}$ are final states).

5. 1. Find ϵ-closure for each state:

- ϵ-closure$(A) = \{A, B, C\}$ (since A has ϵ-transition to B, and B has ϵ-transition to C, plus A itself)
- ϵ-closure$(B) = \{B, C\}$ (since B has ϵ-transition to C, plus B itself)
- ϵ-closure$(C) = \{C\}$

2. Compute transitions without ϵ-transitions ($\delta'(q, a) = \epsilon$-closure$\big(\delta(\epsilon$-closure$(q), a)\big)$):

- **For State A:**
  - $\delta(A, 0) = \{B, C\}$ (from direct transition $A \rightarrow B$ and $A \rightarrow C$)
    - ϵ-closure$(\{B, C\}) = \{B, C\} \cup \epsilon$-closure$(B) \cup \epsilon$-closure$(C) = \{B, C\}$
  - $\delta(A, 1) = \{A\}$ (loop on A)

        - ϵ-closure$(\{A\}) = \{A, B, C\}$
- **For State B:**
    - $\delta(B, 0) = \phi$
        - ϵ-closure$(\phi) = \phi$
    - $\delta(B, 1) = \{B\}$ (loop on B)
        - ϵ-closure$(\{B\}) = \{B, C\}$
- **For State C:**
    - $\delta(C, 0) = \{C\}$ (loop on C)
        - ϵ-closure$(\{C\}) = \{C\}$
    - $\delta(C, 1) = \{C\}$ (loop on C)
        - ϵ-closure$(\{C\}) = \{C\}$

3. Transition Table for NFA without ϵ-transitions:

State	Input 0	Input 1
$\rightarrow A$	$\{B, C\}$	$\{A, B, C\}$
B	ϕ	$\{B, C\}$
$* C$	$\{C\}$	$\{C\}$

- **Initial State:** A
- **Final States:** Since original final state is C, any state whose ϵ-closure contains C becomes a final state. Therefore, **Final States =** $\{A, B, C\}$.